# Trust Without Receipts: Collusion-Aware Social Dispatch and Price-Claim Assessment for Campus Errand Platforms

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***Abstract*—Delivery platforms check a courier's expenses against a menu, a catalogue or a receipt. A campus errand platform has none of these: when a student runner buys a Rs. 20 puff at a hostel canteen, the claim is the only record of the price. We present Errandly, a peer-to-peer campus errand platform, and a problem its design creates: the dispatcher prefers friends, so money circulating inside honest friend groups resembles collusion. We flag only money cycles among mutual friends, then test each flagged group against an acceptance model fitted to logged dispatch rounds from outside every suspected group. Price claims are bound to the lines the requester ordered and judged against a one-vote-per-runner reference held inside an administrator's band. In simulated semesters on a 1,200-student campus that run the production code, with 30 seeds per setting, the cycle search alone reached an F1 score of 0.96. Structure-only and money-only baselines found almost no ring member, and two graph-mining baselines reached at most 0.48. The end-to-end pipeline, cycle search plus policy test, reached 0.93 to 0.97 when rings were careless and 0.87 to 0.88 when they were coordinated. The first version of the test, which reused the router's own scores, incorrectly suppressed 146 of 149 coordinated-ring flags. The price reference stayed at the true price with 60% dishonest runners. We also identify a limit no such test can pass: honest favour-trading and careful collusion leave the same logs. All results come from simulation.**

***Index Terms*—Collusion, fraud detection, peer-to-peer computing, reputation systems, social networks, trust management.**

## I. Introduction

On a residential campus, students already run errands for each other. Someone walking past the canteen brings dinner back to a friend in the hostel; someone going to the main gate collects a parcel for a roommate. Errandly turns this habit into a platform. A student posts an errand with a drop point and a fee, a nearby verified student accepts it, and the requester's money sits in escrow until the handoff is confirmed. The system is a working prototype with an Android and iOS client, a web client and an event-driven backend. It has not yet been piloted with students, so every result in this paper comes from tests and simulations of the code as built.

The hard part is not matching. Commercial delivery platforms control expense fraud by knowing what things cost. Restaurant menus are integrated, and grocery orders are checked against a catalogue or a scanned receipt [1], [2]. A campus canteen offers none of that. It has no point-of-sale integration, no published price list and no receipt for a loose samosa. The runner pays at the counter and types in what it cost, and that number is the only record of the transaction. Neither the requester nor the platform can check it.

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This one property spreads through the whole system. Reimbursement can no longer be verified, only judged for plausibility. Plausibility needs a reference price, which has to be estimated from earlier claims made by the same people the reference will police. An estimator that learns from those claims can be steered by them, which is the situation studied as performative prediction [3]. Judgement then falls on people rather than documents, so reputation carries real weight, and because reputation decides who is offered work, repairing a damaged reputation pays. Repair is easiest among friends. The friendship graph that lets the platform prefer trusted runners is therefore the same structure that makes collusion cheap. Coleman described network closure as the source of the trust and sanctions that hold a group together [4]. On an errand platform, the same closure lets a group coordinate against everyone outside it.

There is a second, quieter problem. Errandly offers each errand to the requester's friends before strangers. Honest friends therefore end up running each other's errands more often than chance would predict, and their money circulates inside the group. A detector that reads in-group circulation as evidence of collusion is then partly reading the router's own output. This is the shape of the runaway feedback loops described for predictive policing [5] and of algorithmic confounding in recommender systems [6]: the system's decisions produce the data that later judges the people affected by them.

This paper is about that interaction between dispatch and detection, and about the price-claim checks that make the platform usable at all. Our contributions are as follows.

1) We formulate *catalogue-free* errand commerce, in which expenditure is unverifiable by construction, and give a threat model in which a successful attack needs no forged record at all (Section III).
2) We show that a social dispatcher and a collusion detector that share one friendship graph interfere, and we measure by how much: staging offers by friendship distance doubles the share of an honest group's errands that stay inside the group, and a cycle detector that ignores this flags honest friend groups (Sections VI-C and VI-G).
3) We introduce *policy-conditioned evidence*. Every dispatch round is logged, a small share of rounds is offered without the social boost, an acceptance model is fitted to logged acceptances by people outside every suspected group, and a group's in-group takes are compared with

what that model predicts through a Poisson-binomial $z$ statistic. We report where this clears honest groups, why the first version, which reused the router's own scores, cleared coordinated rings, and the limit no such test can pass: deliberate in-group routing by honest friends and by a careful ring leave the same logs (Sections IV-F and VI-D).

4) We separate structure from evidence in collusion detection. Neighbourhood closure discounts trust but never accuses anyone; an accusation needs a cycle of qualified payment legs among mutual friends. We compare it with single-signal baselines and with two standard graph-mining baselines, densest-block peeling and label-propagation communities (Section VI-C).
5) We describe price-claim assessment against a reference estimated with one vote per runner inside an administrator's band, with runners who habitually claim just under the limit denied a vote, claims bound to the lines the requester ordered, and shop-level partial pooling. We measure money lost and time to detection under five attacker strategies (Section VI-E).

The platform has not been piloted with students. Every result comes from tests and from simulations that run the production code, and we report the failures we found, several of which led to the code changes described in Section V.

Section II positions this work against prior research. Section III defines the system and threat model, Section IV presents the design and Section V the implementation. Section VI evaluates each mechanism, Section VII discusses limitations, and Section VIII concludes.

## II. Related Work

### A. Crowdsourced Delivery and Spatial Task Assignment

Alnaggar *et al.* survey crowdsourced delivery platforms and classify them by matching mechanism, target market and pay scheme [1]; Le *et al.* review the supply, demand and operations of crowd-shipping [2]. Spatial crowdsourcing studies how location-bound tasks should be assigned to moving workers [7], [8]. These lines of work take the task's price as given. Price anomaly detection in marketplaces, such as the MoatPlus system of Sarpal *et al.*, sets upper price bounds from historical and proximate prices of catalogued items [9]; it assumes the catalogue we lack. Escrow services have been analysed as trusted third parties in online auctions [10], but an escrow can only hold an amount that both sides agree on. In our setting the amount is the thing in dispute.

### B. Trust, Reputation and Sybil Resistance

Reputation systems aggregate past feedback to predict future behaviour [11]–[13], and Hoffman *et al.* catalogue the attacks on them, collusion and identity manipulation among them [14]. EigenTrust propagates trust transitively through a peer network [15], and graph-based trust models for social networks commonly let trust decay with path length [16], which we follow. Sybil defences built on social graphs, such as SybilGuard, rely on a sparse cut between honest and fake regions [17], and Cheng and Friedman show that a large class of reputation functions cannot be Sybil-proof [18]. On our platform identity is tied to an institutional e-mail address, so fake accounts are costly [19]; the attacks that remain are carried out with real accounts. Studies of peer platforms report that ratings are shaped by reciprocity [20], inflate over time [21] and respond to personal trust cues [22]. We take from these that a rating's meaning depends on who gave it.

### C. Closure and Cohesion

Coleman [4] and Granovetter [23] argue that closed, embedded networks support trust and enforcement, while Burt stresses the value of ties that bridge otherwise separate groups [24]. The clustering coefficient of Watts and Strogatz [25] is the usual local measure of cohesion. Community detection methods such as modularity maximisation [26], the Louvain method [27] and label propagation [28] partition a whole graph. The closure score we use is not a new kind of measure. With the student removed, it is a monotone transform, $c = (1-\phi)/(1+\phi)$, of the conductance $\phi$ [29] of the student's friend set, and it is close in spirit to Burt's ego-network constraint [24]. What we need from it is narrow: a per-student number that is cheap to refresh in plain Cypher, easy to explain to a reviewer, and sensitive to whether a neighbourhood is sealed off from the rest of campus. Section VI-B shows that the clustering coefficient does not have the last property. The novelty is in how the score is used, as a discount on trust that never accuses anyone, rather than in the score itself.

### D. Collusion and Group Fraud

Fraud detection on review and social platforms looks for groups that act together: fake reviewer groups [30], spammers with abnormal rating behaviour [31], accounts that act in lockstep [32] and dense blocks hidden behind camouflage edges [33]; Akoglu *et al.* survey graph-based anomaly detection [34]. Shilling-attack detection for recommender systems targets injected profiles and rating time series [35], [36]. Collusion among crowd workers has its own literature. Xu *et al.* review similarity-based, graph-based and learned detectors of worker collusion and note how hard it is to obtain ground truth [37]. Han *et al.* show that collusion can be organised anonymously at large scale and detect it with a graph algorithm over workers' answers [38], and Zhang *et al.* predict fraudulent requesters on a contest platform from their centrality and clustering in the social network [39]. These methods share an assumption: somewhere in the data there is a fake record, a copied answer or an abnormal position in the network. The attack we care about breaks that assumption. A ring's friendships are real, its errands are really run, and the ratings its members leave each other may be perfectly sincere. More importantly for us, none of these detectors runs beside a matcher that itself puts friends together, so none has to separate the platform's influence from the group's.

### E. Truth Discovery and Robust Aggregation

Truth discovery infers true values from conflicting sources by estimating how reliable each source is [40]; multi-truth

discovery allows an object several correct values [41], which matches an item that costs different amounts in different shops. Fang *et al.* show that coordinated workers can poison such aggregates [42]. Robust statistics gives the tools we use: the influence curve [43], the median absolute deviation and its alternatives [44], [45], and practical outlier rules based on it [46]. Partial pooling toward a common mean follows the empirical Bayes argument of Efron and Morris [47].

### F. Feedback Loops and Logged Exploration

Perdomo *et al.* formalise predictions that change the distribution they are evaluated on [3]. Ensign *et al.* show how a predictive-policing system that sends patrols where it previously found crime keeps confirming itself [5], and Chaney *et al.* show how recommender feedback homogenises behaviour [6]. Offline evaluation of deployed systems relies on logged decision probabilities [48] and on a small slice of randomised traffic [49]. Our policy test borrows both ideas in a simple form: the logged scores define a conditional-logit choice model [50], and the number of in-group takes is a sum of independent, non-identical Bernoulli trials [51].

### G. Language Models in a Fraud Pipeline

Text sent to a language model inside an application can carry instructions aimed at the model [52]. In our case every string the model sees is written by the people being judged. We use a locally hosted Qwen2.5 7B model [53] and keep it advisory (Section IV-J).

Table I compares the detectors closest to ours on the three properties that matter here.

## III. System and Threat Model

### A. Actors and Errand Lifecycle

Students register with an institutional e-mail address verified by a one-time code. Any student can post an errand as a *requester* or accept one as a *runner*; administrators maintain reference prices and review flags. An errand has a drop point, a fee, an optional amount of cash to collect (for example at the gate) and an item list. Items are of three kinds: menu items from vendors registered on the platform, packaged goods with a printed maximum retail price (MRP), and *loose-priced* items such as canteen food that carry no printed price. Only the last kind is reimbursed on the runner's word.

Fig. 1 shows the lifecycle. Money is placed in escrow before any runner sees the errand. Offers go out in timed tiers (Section IV-H). Acceptance takes a Redis lock and then a row lock, so exactly one runner wins a race. Before marking the errand picked up, the runner must report the price of every loose-priced line, and each line is judged at that moment. Settlement happens once, when the requester confirms completion.

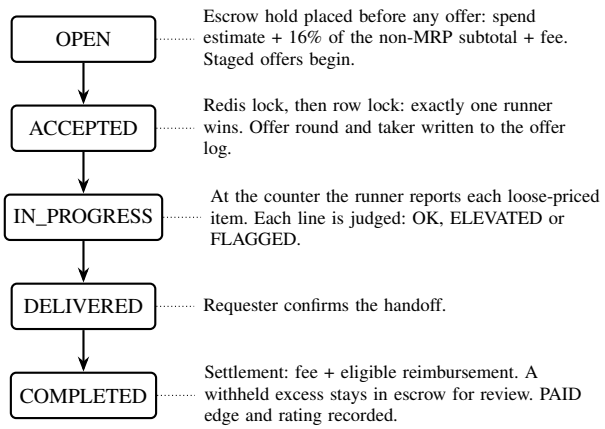



Fig. 1. Errand lifecycle. Money is held before any offer goes out, every loose-priced line is judged at the counter, and settlement pays only the eligible amount.

### B. Graphs, Logs and Claims

Let $U$ be the students on one campus. The *friendship graph* $G_F = (U, E_F)$ is undirected; an edge exists only for an accepted friend request and records when it was accepted. We write $N(u)$ for the friends of $u$.

The *payment graph* $G_P$ is a directed multigraph. A completed errand $e$ posted by $a$ and run by $b$ adds an edge $a \to b$ with time $t_e$ and amount $x_e$, the fee plus any cash the runner collected for $a$. The reimbursement for goods is deliberately left out. It repays a shop for things $a$ received, so it is not value transferred from $a$ to $b$, and counting it would make a large grocery run for a friend look like a large gift.

The *offer log* records, for every dispatch round, the requester, each candidate's distance, friendship distance, trust, reputation, penalty and resulting score, whether the round was exploratory, and who accepted.

An errand's *order lines* are what the requester asked for: a name, a quantity and, for an item picked from the administrator's price list, a link to its reference. A *claim* is the runner's report of the unit price paid for one order line. The unit is one of whatever the line names (one puff, one bottle), and the claimed quantity may not exceed the quantity ordered. A claim records the runner, item, shop, unit price, quantity, the reference and allowance it was judged against, and the verdict.

*Reference notation.* For item $k$, $r_k$ is the campus reference in force. The administrator sets its initial value and a band $[L_k, U_k]$; $\hat{r}_k$ is the estimate computed from recent claims, which replaces $r_k$ only inside the band (Section IV-C); and $r^*_{k,s}$ is the shop-adjusted value used to judge a claim made at shop $s$ (equation (2)). Where the shop is unknown or barely observed, $r^*_{k,s} = r_k$.

*Outcome terms.* We use five words with fixed meanings. A claim, a runner or a group is *flagged* when a rule writes a flag record; a flag is evidence, not a verdict. A ring flag is *suppressed* when the policy test (Section IV-F) finds the group's in-group activity explained, in which case no flag is written. A flag is *reviewed* when an administrator marks it upheld or dismissed. A student is *sanctioned* when an action is applied to their account: a warning, a rank penalty, a pause on running or a suspension. A student is *cleared* when a dismissal releases withheld money and lifts any pause that the remaining flags no longer justify (Section IV-B).

TABLE I
COLLUSION AND FRAUD DETECTORS CLOSEST TO THIS WORK

| Work | Setting | Evidence used | Needs a fake record | Accounts for the matcher | Judges unpriced claims |
|---|---|---|---|---|---|
| Mukherjee *et al.* [30] | product reviews | group behaviour features | yes | no | no |
| CopyCatch [32], FRAUDAR [33] | social and e-commerce graphs | lockstep edges, dense blocks | yes | no | no |
| Han *et al.* [38]; review [37] | crowdsourced labelling | answer similarity, worker graphs | yes | no | no |
| Zhang *et al.* [39] | contest crowdsourcing | centrality and clustering of requesters | no | no | no |
| Fang *et al.* [42] | truth discovery | source reliability | yes | no | no |
| MoatPlus [9] | online marketplace | catalogue price history | no | no | catalogued only |
| Ensign *et al.* [5] | predictive policing | (analysis of a feedback loop) | no | yes | no |
| **This work** | campus errands | money cycles among friends, tested against a fitted model of logged routing | no | yes | yes |

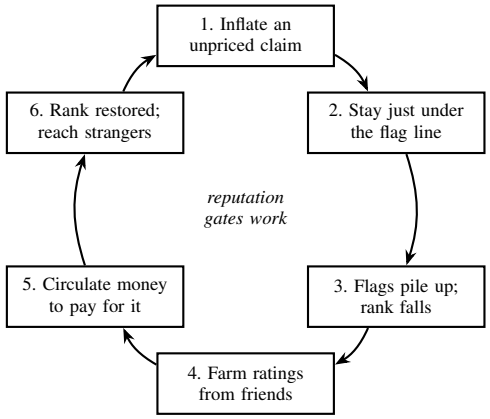



Fig. 2. The attacker's loop. Each stage is answered by a control in Table II.

TABLE II
ATTACK STAGES AND THE CONTROLS THAT ANSWER THEM

| Stage | Attacker move | Control (Section) |
|---|---|---|
| 1 | inflate the price of a loose item | reference with band; withhold excess (IV-B, IV-C) |
| 2 | keep every claim just under the limit | distribution rule; no vote on the reference (IV-B) |
| 3 | absorb flags; rank falls | strike ladder; capped rank penalty (IV-B, IV-H) |
| 4 | farm ratings from friends | one vote per rater; concentration discount (IV-G) |
| 5 | circulate money to fund stage 4 | cycles among friends; policy test (IV-E, IV-F) |
| 6 | make farmed errands look real | language channel, advisory only (IV-J) |

### C. Threat Model

The attacker is a verified student who wants to be paid more than the work is worth, acting alone or with genuine friends. We assume the attacker knows every rule in this paper, can coordinate with friends over chat, and can run real errands at real cost. We assume the attacker cannot create accounts in bulk, because each needs an institutional address, and cannot tamper with the server or its logs.

Fig. 2 and Table II show the campaign we design against. The loop closes because reputation decides who is offered work: a runner whose rank has fallen needs to repair it before reaching new victims, so repairing it is a precondition of the fraud rather than vanity. Stages 4 and 5 need no forged data. The friends are real, the errands are run, and the five-star ratings may be sincere.

Out of scope are a shopkeeper who quotes a false price on the runner's behalf, shared accounts, coercion, a compromised administrator and spoofed device locations.

## IV. DESIGN

### A. Overview

Fig. 3 shows the architecture. PostgreSQL is the only source of truth. Every state change writes a domain event to an outbox table in the same transaction, a relay publishes it to Kafka, and four consumer groups handle notifications, analytics, settlement and the projection of friendships and payments into Neo4j. The graph is a derived read model: nothing on a request path writes to it, it can be rebuilt from the event log, and every read from it sits behind a circuit breaker. A scheduler escalates offers, refreshes graph metrics and runs the fraud sweeps. Table III collects the parameters introduced in the rest of this section.

### B. Assessing a Price Claim

A claim must first be tied to the right reference, because a claim that matches no reference is never checked at all. On an itemised order the claim is bound to one of the order's own lines. Its name is matched against the names of those lines, exactly after normalisation and otherwise by fuzzy matching. A line the requester picked from the administrator's price list is then judged against that line's reference, whatever the runner calls it. A name that matches no line is refused, and so is a quantity above the quantity ordered. Before this rule, a runner could rename a priced item ("zzq special") until it matched nothing and be paid unchecked; Section VI-E measures the difference. On an errand with no itemised lines the claimed name is mapped to a campus-wide key instead. Deterministic rules handle case, punctuation, plurals and common abbreviations ("chkn puf" becomes "chicken puff"), and fuzzy matching with a similarity threshold of 0.86 handles typos against the keys a campus already prices. A language model may *suggest* aliases for names that recur unmatched, but nothing is judged against an alias until an administrator approves it.

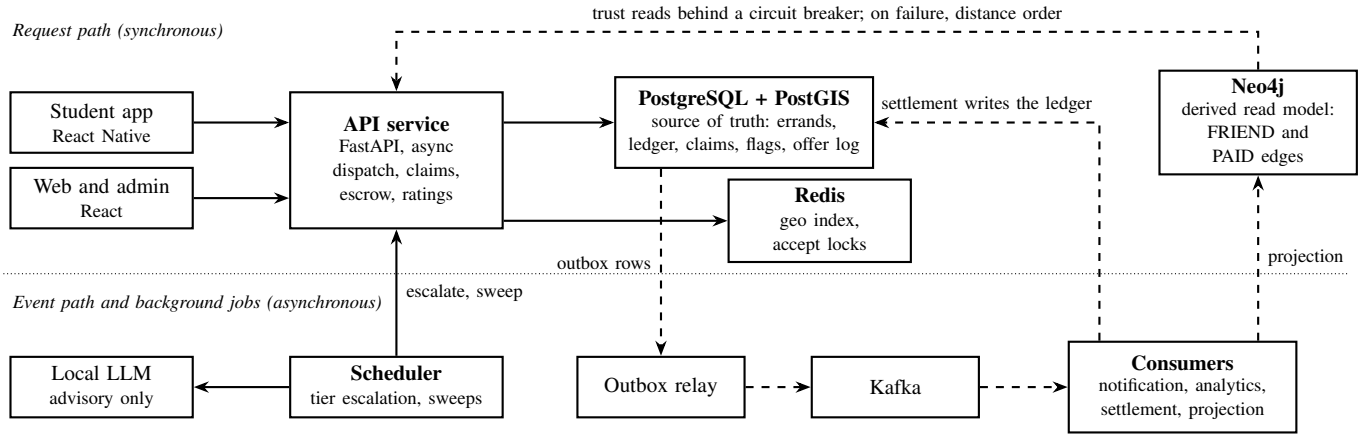



Fig. 3. Architecture in two lanes. The upper lane is the request path, the lower lane the event path and background jobs. Solid arrows are synchronous calls; dashed arrows are events or read-model traffic. Losing Neo4j degrades matching to distance order and never blocks a request.

For item $k$ with reference price $r_k$, the allowance is

$$\tau_k = \min\big(\max(0.40\, r_k,\ Rs.\,5),\ Rs.\,20\big), \qquad (1)$$

where the percentage, floor and cap are per-item settings. A flat allowance would be 80% of a Rs. 25 puff and 4% of a Rs. 500 grocery run; scaling it protects the cheap items that make up most of the traffic. A claimed unit price $c$ at shop $s$, judged against $r^*_{k,s}$ (Section IV-C), is *FLAGGED* if $c - r^*_{k,s} > \tau_k$, *ELEVATED* if $0 < c - r^*_{k,s} \le \tau_k$ and *OK* otherwise. For a line of quantity $q$, a flagged claim is reimbursed at $q\, r^*_{k,s}$ and the excess stays in escrow for review; the other two are paid $q\, c$ in full. An item with no reference (the cold-start case, Section IV-C) is paid as claimed, up to the escrow hold, and becomes evidence for one.

One high claim is not fraud; prices do rise. Sanctions therefore key on a pattern, and only price-claim flags count toward it; ring and rating-farming flags never feed the ladder, because they are relational inferences that a person has to weigh. With 3, 5 and 8 flagged errands within 30 days the system applies, automatically, a warning, a 0.5-star reputation penalty and a seven-day pause on running. These three steps are temporary and reversible. The fourth, account suspension, needs 12 flags that an administrator has *upheld*; open flags alone stop at the pause. Flags are counted per errand, so one receipt with four inflated lines is one incident. Every sanction notifies the student with the reason and a route to an administrator. That is the appeal path. When an administrator dismisses a flag, the withheld money is paid to the runner, the claim is restored as evidence, and a running pause that the remaining flags no longer justify is lifted. The strike stays on record, marked as lifted.

A runner can also sit just under the limit on every claim. A claim is *near the line* if $c - r^*_{k,s} \ge 0.6\, \tau_k$. A runner with at least four near-line claims in 30 days, forming at least 60% of their judged claims, receives a `PERSISTENT_NEAR_THRESHOLD` flag. Requiring a share as well as a count keeps a busy honest runner, whose elevated claims are diluted by ordinary ones, out of it. No money is withheld for this flag, because no single claim broke a rule.

### C. Estimating References From Adversarial Claims

The administrator sets, for each loose-priced item, a reference and a band $[L_k, U_k]$. An hourly job re-estimates the reference from the last 30 days of claims. Flagged claims are excluded, and so are all claims by runners under an open or upheld near-line flag. Their claims are individually legal, which is why they would otherwise drag the line upward. Claims are grouped by runner and each runner contributes one vote, the median of their own claims $m_i$. With $M = \mathrm{med}_i\, m_i$ and $\mathrm{MAD} = \mathrm{med}_i\, |m_i - M|$, votes with $|m_i - M| > 3 \times 1.4826 \times \mathrm{MAD}$ are dropped and the estimate is the median of the rest [44], [46]. At least three runners and five claims are needed; with fewer, or when every vote is rejected, $r_k$ is left as it was. The estimate $\hat{r}_k$ replaces $r_k$ only if it lies inside $[L_k, U_k]$; outside the band it becomes a proposal for an administrator, and within 5% of either edge it is applied and also proposed. The band is never moved by the data it bounds.

*Cold start.* A new item has no reference until an administrator lists it with a price and band, and until then its claims are paid as claimed, up to the hold. The hold is the only bound in this state: a claim that pushes the payout above the hold freezes it for review (Section IV-I). Unmatched names are queued for alias review so that a recurring item is noticed. A listed item starts at the administrator's price, $\hat{r}_k$ takes over once three runners have reported it, and shop pooling begins only once three distinct runners have reported from that shop.

Without the vote limit, one runner who claims Rs. 40 two hundred times would set the reference. Without the band and the exclusions, a performative loop forms [3]: inflated claims raise the reference, a higher reference widens the allowance in (1), and the next inflated claim looks normal.

An item also has different prices in different shops [41]. A single campus reference sits between them and is wrong both ways: the honest runner at the dearer counter is always elevated, and an inflater at the cheaper one blends in. For shop $s$ with $n_s \ge 3$ distinct reporting runners and median of per-runner medians $M_s$, the reference used for judging is pulled toward the shop:

$$r^*_{k,s} = w_s M_s + (1 - w_s)\, r_k, \qquad w_s = \frac{n_s}{n_s + 6}, \qquad (2)$$

capped at $2.5\,r_k$ as a sanity bound [47]. The allowance still scales with the campus reference $r_k$, so a shop cannot widen its own allowance by being dear. This step deliberately does not exclude flagged claims: excluding them would deadlock a genuinely dearer shop, whose honest claims are all flagged against the campus reference. Independence, counted in distinct runners, is what protects it instead.

### D. Trust From the Friendship Graph

Trust from requester $q$ to candidate runner $v$ at friendship distance $h \le 4$ is

$$T(q,v) = \delta^{\,h-1}\,\big(1-\pi\big)\,\mu, \qquad \delta = 0.45, \tag{3}$$

where $\mu \in [0.4, 1]$ grows linearly to 1 over the first 30 days of the newest friendship on the path, so that a network recruited yesterday counts for less than an established one. Old friendships are not discounted.

The penalty $\pi$ measures how sealed off the neighbourhood carrying the trust is. For a student $x$ with $|N(x)| \ge 2$ we count edges among $x$'s friends, $I(x)$, and edges from $x$'s friends to anyone outside $N(x) \cup \{x\}$, $B(x)$, and define *closure*

$$c(x) = \frac{I(x)}{I(x) + B(x)}. \tag{4}$$

When $|N(x)| < 2$ or $I(x) + B(x) = 0$, as for a student whose friends have no other ties at all, $c(x) = 0$: there is no neighbourhood to call sealed. Two choices matter. First, $x$ itself is excluded; counting $x$'s own spokes as internal edges makes an open star look closed. Second, unlike the clustering coefficient $2I(x)/(k(k-1))$ with $k = |N(x)|$ [25], closure counts the ties that leave the neighbourhood. Three friends who know each other but each have twenty friendships elsewhere form a maximally clustered but open neighbourhood (Fig. 4(c)).

The structural penalty rises linearly from 0 at $c = 0.55$ to a cap of 0.70 at $c = 1$. For a path through intermediary $x$, it is raised toward 0.95 in proportion to money evidence $g(\kappa_x)$ (Section IV-E):

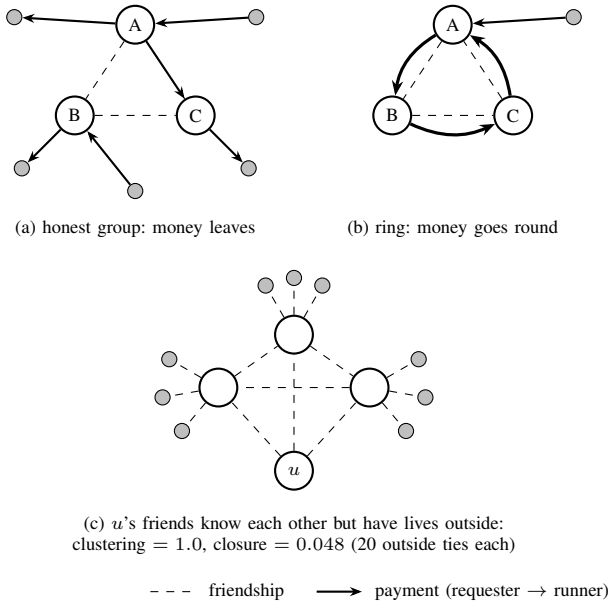
$$\pi_{\text{via}} = s(c_x) + \big(0.95 - s(c_x)\big)\, g(\kappa_x), \tag{5}$$

where $s(\cdot)$ is the structural part. A direct friend has no intermediary to judge, and structure alone never discounts a direct friend: a close friend group is the strongest honest signal the platform has. A direct friend is discounted only by $\pi_{\text{dir}} = \min(0.95,\ 0.95\,c_v\,g(\kappa_v))$, that is, only when money corroborates the shape. No penalty reaches 1, so nobody disappears from matching on a suspicion.

### E. From Capacity to Evidence: Money Cycles Among Friends

Structure cannot separate three roommates who take turns fetching dinner from three students farming ratings off each other: the two subgraphs are isomorphic (Fig. 4(a), (b)). What differs is whether value circulates.

The *circulation* $\kappa_u$ of a student is the share of the value they moved on the platform in the last 180 days, in either direction, that moved between them and their own friends. It is set to 0 below Rs. 300 or four payments, so a newcomer whose first




(a) honest group: money leaves

(b) ring: money goes round

(c) $u$'s friends know each other but have lives outside: clustering $= 1.0$, closure $= 0.048$ (20 outside ties each)

Fig. 4. (a) and (b) are isomorphic friendship triangles; only the payments differ. (c) A neighbourhood the clustering coefficient scores as fully cohesive but that is open to the rest of campus.

two errands were for a friend is not read as wholly internal. The corroboration term in (5) is $g(\kappa) = \min\big(1, (\kappa - 0.6)/0.4\big)$ for $\kappa > 0.6$ and 0 otherwise. Circulation only discounts trust; it accuses nobody.

An accusation needs a closed money cycle. For every ordered pair of mutual friends $(a, b)$ we total the value $V_{ab}$ and count the distinct errands $n_{ab}$ on edges $a \to b$ in the window. A leg qualifies if

$$n_{ab} \ge 2 \quad \text{and} \quad \big(V_{ab} \ge Rs.\,150 \text{ or } n_{ab} \ge 5\big). \tag{6}$$

The frequency branch closes a loophole: fees may be zero, and a ring that cycles zero-rupee errands still buys completed errands and ratings. Qualifying legs form a directed graph, and every strongly connected component with at least three members, found by an iterative form of Tarjan's algorithm [54], is a ring of any size. Every such component is reported; an earlier version returned only the 50 of highest value, which on a busy campus would have hidden the rest until the larger ones aged out. Filtering legs before the search matters: if a minimum were taken over a whole component afterwards, a ring could hide behind one deliberate Rs. 1 payment. Every member of a ring is flagged, because a cycle is symmetric and names no ringleader. No money is withheld, because the cycle identifies a group, not a dishonest errand.

Why is a cycle suspicious at all, when the platform injects no money and a pure cycle is zero-sum? Because the motive is cost avoidance. Farming ratings needs completed errands, each paid for by a confederate. When $A$ pays $B$, $B$ pays $C$ and $C$ pays $A$, each has paid out about what they received, and the group has produced three completed errands and three ratings at almost no net cost. The money taken from strangers comes later, through the rank those ratings buy. An honest friend group's money leaks out to the rest of campus, and for them an internal errand remains a real expense.

### F. Separating the Router's Influence From the Group's Choice

Because dispatch offers errands to friends first, an honest close group produces money cycles too. Before a ring flag is raised, we ask how much of the group's in-group activity the routing itself predicts (Fig. 5).

*The acceptance model.* For each logged round $j$ with candidate set $K_j$ ($|K_j| \geq 2$), the chance that candidate $k$ is the one who accepts is a conditional logit [50]:

$$p_{jk} = \frac{\exp(\beta^\top f_{jk})}{\sum_{l \in K_j} \exp(\beta^\top f_{jl})}, \tag{7}$$

where $f_{jk}$ holds four logged features: distance in kilometres, whether $k$ is a direct friend of the requester, whether $k$ is a friend of a friend, and $k$'s reputation above the neutral 3.5. The weights $\beta$ are fitted by maximum likelihood (Newton's method with a small ridge term, $10^{-2}$) on every round in the window whose acceptor was among its candidates, *excluding rounds posted by a member of any candidate ring*, so a group never helps calibrate the model that judges it. With fewer than 200 such rounds the model is not trusted, and the test falls back to the router's own logged scores $e_{jk}$, using $p_{jk} \propto \exp(-e_{jk}/\theta)$ with $\theta = 300$ m. That fallback was the original design, and Section VI-D shows why it failed.

*The test.* For the rounds posted by members of the group $S$, let $\rho_j = \sum_{k \in K_j \cap S} p_{jk}$ be the predicted chance that a member took round $j$, and let $O$ count the rounds a member actually took. Treating rounds as independent trials with different probabilities, the count is Poisson-binomial [51], with expectation $E = \sum_j \rho_j$ and variance $V = \sum_j \rho_j(1-\rho_j)$, and

$$z = \frac{O - E}{\sqrt{V}}. \tag{8}$$

If $V$ is numerically zero, every round was a foregone conclusion: $z = 0$ when $O$ matches $E$, and otherwise $z$ is recorded as $\pm 99$, which is never explained. With at least 25 usable rounds in 180 days and $z < 3$, the routing explains the group's behaviour and the ring flag is suppressed. Otherwise, or with too little history, the flag stands and goes to a reviewer. The test can only clear; a high $z$ never raises a severity or creates a flag.

*What the test can and cannot identify.* The model describes how students *outside* suspected groups respond to being offered an errand by a friend at a given distance. A ring is separable from honest friends only to the extent that it takes its own errands more often than such friends do. An honest group that deliberately routes favours to one another breaks that assumption in exactly the way a ring does, so no test built on these logs can tell the two apart; only the content of the errands, or a person, can. We return to this limit in Section VII.

A test like this is only informative if some rounds are not shaped by the social boost. In 5% of rounds the first offer is *exploratory*: it goes to the five nearest runners with no trust term and no friendship filter, and its scores are logged without trust. This costs those requesters a less socially matched runner. It is paid to keep a control group in the data [48], [49].

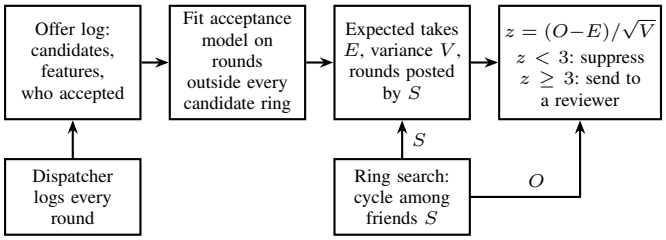



Fig. 5. Policy-conditioned evidence, read left to right. An acceptance model fitted on rounds posted outside every candidate ring predicts how often the flagged group $S$ should have taken its own errands; only the unexplained excess is evidence.

### G. Reputation That Counts People, Not Ratings

Errandly keeps two numbers per runner. The plain average is what a student sees on a profile. Ranking uses a provenance-weighted score. For runner $v$, let the *concentration* $\phi$ be the share of $v$'s ratings given by $v$'s friends. Each distinct rater $i$ casts one vote $\bar{r}_i$, the mean of their ratings, with weight

$$w_i = \begin{cases} 1 - \min\big(0.75,\ 1.5\,(\phi - 0.5)\big), & i \in N(v),\ \phi > 0.5, \\ 1, & \text{otherwise.} \end{cases} \tag{9}$$

The score $\bar{s} = \sum_i w_i \bar{r}_i / \sum_i w_i$ is shrunk toward a neutral 3.5 by the effective sample size $W = \sum_i w_i$:

$$m = \frac{W}{W+8}\,\bar{s} + \frac{8}{W+8} \times 3.5. \tag{10}$$

The discount attaches to concentration, never to friendship: twenty strangers and five friends leave $v$ untouched. A reputation carried by $v$'s own circle becomes low-confidence rather than low, and climbing back requires strangers. Counting people instead of ratings removes the patience exploit; under per-rating weights, 100 farmed ratings from friends outscored 20 honest ones from strangers.

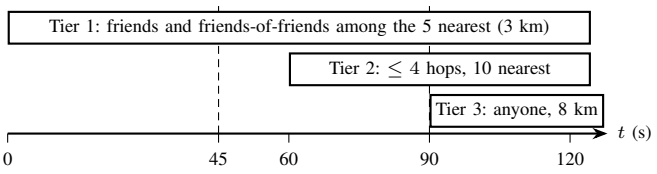
Three detectors raise a `RATING_FARMING` flag for review once a runner has at least four in-cluster ratings and a concentration of at least 0.70: friends rating $v$ at least 0.8 stars higher than at least three strangers do; at least two high in-cluster ratings within 14 days of a penalty; or at least 15 in-cluster ratings with at most one from outside and at least 2.5 ratings per friend, the pattern of a runner who builds standing inside a circle before meeting a victim.

### H. Dispatch: Staging Offers in Time

Candidates come from a Redis geospatial index. Each is scored in metres of *effective distance*, so the four terms can be compared directly:

$$e = d - 1500\,T - 800\,(m - 3.5) + P, \tag{11}$$

where $d$ is distance to the drop point, $T$ is trust from (3), $m$ the weighted reputation and $P$ an integrity penalty for open, unreviewed flags: 300, 700 or 1200 m by severity, decaying linearly over 30 days and capped at 2000 m. The cap is deliberate. Without it, open flags would add up until the penalty exceeded any plausible distance, turning a demotion into a ban that no human decided. An errand is also never




Tiers fire on the first 30 s scheduler sweep after 45 s and after 90 s (shown here at 60 s and 90 s). In 5% of rounds the first offer is socially blind: all 5 nearest runners at once, with no trust term.

Fig. 6. Staged dispatch. Social preference comes from when each circle is told, not from the order of a single broadcast.

offered to a runner who shares an open ring flag with that requester.

Sorting is not enough. Every candidate in a round is notified within milliseconds, so ordering a simultaneous broadcast leaves a race that the nearest stranger often wins. Errandly stages offers in time instead (Fig. 6). The first round goes to the runners among the five nearest within 3 km who are within two friendship hops. The scheduler, sweeping every 30 s, widens the offer after 45 s to four hops among the ten nearest and after 90 s to anyone among the ten nearest within 8 km. If nobody within two hops is near when the errand is posted, the first round goes straight to the five nearest, because an errand nobody sees is worse than one a stranger takes, and the 90 s widening to 8 km still applies to it. (In the first version it did not, and such errands expired more often; Section VI-G.) When the graph cannot answer, every function returns `None` rather than an empty set, and `None` filters out nobody.

### I. Escrow With Targeted Headroom

A runner who fronts cash at the counter relies on the hold existing, so it is placed before any offer:

$$H = X + C + 0.16\, X_{\text{loose}} + F, \tag{12}$$

where $X$ is the priced item total, $C$ the cash to collect, $X_{\text{loose}}$ the subtotal of loose-priced items at their references and $F$ the fee. Only the loose-priced part gets headroom, because a menu price or a printed MRP is already known. At settlement the runner is paid the fee and the eligible reimbursement; any surplus returns to the requester at once. If what the runner is owed exceeds the remaining hold, the payout is frozen for an administrator, because paying in full would charge the requester more than they agreed to lock, and paying the capped amount would make the runner absorb a gap that may be a genuine price rise. Balances are never stored; a balance is a sum over an append-only ledger, and a uniqueness constraint on (errand, user, entry type) makes a redelivered settlement event a no-op.

### J. The Language Channel

Structure and money answer *who* and *how much*; neither answers *what for*. When a ring flag is raised, a locally hosted model reads the errand texts of the group and returns a schema-constrained assessment for the reviewer. The channel is advisory: it never raises a severity, never withholds money and never creates a flag. Its useful direction is exculpatory, where being wrong costs least. All text is fenced and labelled as untrusted, and the output schema leaves a crafted note nothing to do but move a number a human can see [52]. A second channel that reads rating comments is switched off, for the reason given in Section VI-H.

TABLE III
MAIN PARAMETERS OF THE DEPLOYED SYSTEM

| Parameter | Value |
|---|---|
| Allowance on a loose item | 40% of reference, Rs. 5 to Rs. 20 |
| Near-line rule | $\geq 0.6\,\tau$; $\geq 4$ claims and $\geq 60\%$ |
| Strike ladder (flagged errands, 30 days) | 3 / 5 / 8 open; 12 upheld |
| Reference estimate | $\geq 3$ runners, $\geq 5$ claims; cut at $3 \times 1.4826$ MAD |
| Shop pooling | prior 6; $\geq 3$ runners |
| Trust hop decay $\delta$; maximum hops | 0.45; 4 |
| Friendship maturity | 30 days; floor 0.40 |
| Closure knee; structural / corroborated cap | 0.55; 0.70 / 0.95 |
| Circulation knee; evidence floor | 0.60; Rs. 300 and 4 payments |
| Money window | 180 days |
| Ring leg | $\geq 2$ errands and ($\geq$Rs. 150 or $\geq 5$ errands) |
| Policy test | $\geq 25$ rounds; $z < 3$ clears |
| Acceptance model | $\geq 200$ rounds; ridge $10^{-2}$; else $\theta = 300$ m |
| Exploration rate | 5% of first offers |
| Trust / reputation weight in (11) | 1500 m / 800 m per star |
| Open-flag penalty | 300 / 700 / 1200 m; cap 2000 m |
| Rating concentration knee; maximum discount | 0.50; 0.75 |
| Rating confidence prior | 8 ratings at 3.5 |
| Escrow headroom | 16% of loose-priced subtotal |

## V. IMPLEMENTATION

The backend is about 12,900 lines of Python 3.12 using FastAPI and SQLAlchemy 2.0 (asynchronous), with PostgreSQL 16 and PostGIS for errands, ledger and fraud data, Redis 7 for the runner geospatial index, accept locks and publish–subscribe, Kafka 3.8 in KRaft mode as the event backbone, Neo4j 5.26 Community Edition for the friendship and payment graph, and MongoDB 7 for chat. Twenty-four migrations define the schema and the API exposes 70 HTTP routes plus WebSocket channels for live tracking and offers. The student client is built with React Native 0.86 and Expo 57, and the web client with React 18, TypeScript and Leaflet. The language channel runs Qwen2.5 7B through Ollama on campus hardware, so no student text leaves the institution.

Money moves in exactly one place: the settlement consumer, on the completion event. Two independent gates make Kafka's at-least-once delivery harmless: a processed-events table at the consumer and the ledger's uniqueness constraint. This follows the transactional outbox pattern [55], [56]. Graph reads use circuit breakers [57]; a graph outage degrades matching to distance order. Graph metrics are computed in plain Cypher; we avoided the Graph Data Science plugin, whose download at container start would make an offline start fail.

We treat explainability as a design rule. Every flag stores the evidence that raised it: the reference and allowance a claim was held to, the shop and how many runners have priced it, a ring's members, laps, value and policy statistic. Every offer round stores the four terms of (11) for every candidate. These decisions withhold money and suspend students, and an accusation that cannot be explained cannot be contested [58]. This is also why learned models are confined to the advisory channel.

*Changes made after the first evaluation.* The first round of experiments exposed six faults, and each was fixed in the production code, with a regression test, before the results reported here were produced. (i) A claim could be renamed until it matched no reference and was then paid unchecked; claims are now bound to the order's own lines, and quantities are capped at the quantity ordered. (ii) The policy test reused the router's scores as its model of acceptance and cleared coordinated rings; it now fits an acceptance model to logged acceptances. (iii) When the predicted probabilities left no variance, the test reported $z = 0$ and cleared the group even when the group had taken rounds it had no predicted chance of taking. (iv) Errands whose first offer went out unrestricted were never widened. (v) The ring search returned at most 50 rings. (vi) Unreviewed ring flags counted toward the automatic strike ladder, and its last step, account suspension, needed no human. Section VI reports the original behaviour beside the fixed one where the difference is measurable.

The backend has 273 automated tests (6,014 lines). In an isolated container against its own database, a separate Redis database and a separate Neo4j instance, all 273 passed in 111 s.

## VI. Evaluation

### A. Method

There is no pilot data yet, so we evaluate in simulation, and we take care over what is simulated. The experiment scripts import the production modules unchanged. Only storage is replaced: database sessions are swapped for small fakes that return prepared rows, and graph queries run against a separate, disposable Neo4j instance loaded with the simulated campus. The decision logic, constants and Cypher queries are the deployed ones. Every number below is regenerated by one script per experiment in the repository, and every result file records the commit it was produced from, the seeds and the machine.

*Seeds and intervals.* The collusion, price-claim, dispatch and escrow experiments use 30 seeds for their main results and 10 seeds per point for parameter sweeps. We report means with the half-width of a two-sided 95% confidence interval of the mean, from the Student $t$ distribution across seeds. A seed fixes the friendship graph, the planted groups, the homes, who is online, every runner's willingness and delay, and the exploration coin, so two policies run on the same seed see the same semester.

*Hardware.* All experiments ran in Docker containers on one laptop with an Intel Core i7-12650H (10 cores, 16 threads) and 7.6 GiB of memory available to Docker, with Python 3.12 and Neo4j 5.26 Community Edition. The timing experiment (Section VI-H) ran alone on the machine; the others ran in parallel, each against its own Neo4j instance.

The campus has 1,200 students in 24 hostel blocks of 50. Friendships follow a stochastic block model [59], with probability 0.10 within a block and 0.0015 across blocks, giving a mean degree of 7.2; residence-dominated community structure of this kind has been observed in real campus networks [60]. We plant 28 honest close-knit groups (sizes 3 to 8, 120 students) and five collusion rings (sizes 3, 4, 5, 6 and 8, 26 students), each as a clique inside one block, so honest groups and rings are structurally identical by construction. Over a 180-day semester each student posts errands at 0.12 per day; fees are drawn from Rs. 20–50, and 30% of errands carry Rs. 50–400 of cash to collect. Each errand sees a random 12% of students online. Dispatch uses the production ranking, tiers, scheduler cadence and exploration coin. An offered runner accepts with probability $0.5\,e^{-d/2000\,\mathrm{m}}$, multiplied by a friend affinity of 1.5 for direct friends, after a log-normal delay with median 20 s. These behavioural parameters are assumptions, which is why we vary them. Ring members each post 12 extra low-fee errands that a ring partner takes from the feed within 3–15 s.

To check that the simulator dispatched as production would, closure and circulation computed inside the simulator were compared with the production Cypher on the final graph of every trial; the largest difference in any trial was 0.

Table IV collects the main results; the subsections that follow give the detail behind each row.

### B. Closure Versus Clustering

Table V scores canonical neighbourhoods with the production `REFRESH_METRICS` query on Neo4j and, as a cross-check, with an exact Python transcription of it; the two agreed on all 425 nodes.

The two measures agree more often than one might expect. On a sealed clique both are 1.0 at every size, and when each clique member has one private friend they give identical values: 0.50 for four members and 0.80 for ten. Closure is therefore not size-invariant in general, and we do not claim that it is. Where the measures part ways is a neighbourhood whose members have lives outside it. When ego's three friends all know each other but each has twenty friendships elsewhere, the clustering coefficient is 1.0 and, used in place of closure, would apply the full 0.70 structural discount. Closure is 0.048 and applies none. In the second-to-last row, counting ego's own spokes as internal edges would lift the closure of an open hub from 0 to 0.50, which is why the query excludes ego.

### C. Detecting Rings Among Honest Friend Groups

*Regimes.* We compare ten detectors in five behavioural regimes, 30 seeds each. In the *routine* regime friends meet only through the router. The *busy* regime has twice as many runners online and friends four times likelier to accept each other's offers, so the router itself puts honest friends together. In the *favours* regime every honest group member also posts six errands per semester meant for a particular friend, who takes them from the feed; this is deliberate in-group routing

TABLE IV
SUMMARY OF RESULTS. MEAN (95% CI HALF-WIDTH) UNLESS STATED

| Area | Measure | Result | Runs (script) |
|---|---|---|---|
| Ring detection, deployed pipeline | F1, routine / busy / favours / coordinated / hard | 0.97 (0.02) / 0.96 (0.02) / 0.93 (0.03) / 0.88 (0.04) / 0.87 (0.04) | 30 seeds (exp2) |
| | Ring members found (TP / FN of 26), same order | 24.7/1.3, 24.9/1.1, 24.4/1.6, 20.7/5.3, 22.1/3.9 | |
| | Cycle search alone (base), F1 routine / coordinated | 0.96 (0.02) / 0.99 (0.01) | |
| | Best graph-mining baseline, F1 | 0.28 to 0.48 (FRAUDAR-style, given the true count) | |
| Honest impact | Honest students flagged per trial (of 1,174), same order | 0.2 (0.2), 0.6 (0.4), 1.9 (1.0), 0.0 (0.1), 2.3 (1.0); FPR at most 0.20% | 30 seeds (exp2) |
| | With 24 honest favours per member | 110 per trial: the identifiability limit | 10 seeds (exp2) |
| Policy test | Coordinated-ring recall, router scores vs fitted model | 0.02 (0.02) vs 0.79 (0.06) | 30 seeds (exp2) |
| | Honest-only cycles cleared, busy / favours / hard | 25 of 26, 35 of 50, 28 of 48 | |
| | Adaptive ring, members caught at 4 / 8 / 12 farmed errands | 0% / 42% / 86% | 10 seeds (exp2) |
| Price claims | Reference after 16 weeks, any strategy, up to 60% dishonest | Rs. 20.00 in every seed (true price Rs. 20) | 30 seeds (exp3) |
| | Honest claims flagged; honest near-line flags | 0 of all honest claims; 0 runners | 4,620 runs (exp3) |
| | Line-walkers at 30%: days to flag; excess before flag; 16 weeks unreviewed | 19.3 (1.2) days; Rs. 331 (4); Rs. 1,958 (46) | 30 seeds (exp3) |
| | Evasive at 30%: share flagged; days to flag | 79%; 46.4 (3.2) days | |
| | Adversarial names paid unchecked, old rule vs deployed | 41 of 61 vs 6 of 61 (all swaps to an unpriced line) | deterministic (exp9) |
| | Colluders needed to move a shop with 0 / 10 honest reporters | 3 / 10 | deterministic (exp4) |
| Dispatch | Errands taken within 2 hops: broadcast / staged / one at a time | 17.8% / 35.8% / 50.4% (all $\pm 0.3$) | 30 seeds (exp6) |
| | Median wait to accept, same order | 12.3 / 19.0 / 41.9 s | |
| | Unaccepted after 30 min, staged, original vs fixed scheduler | 1.88% (0.03) vs 0.22% (0.01) | |
| Escrow | Payouts frozen at 16% headroom; money locked without need | 9.0% (0.2); Rs. 12.7 (0.1) per order | 30 seeds (exp7) |
| Runtime | Three graph jobs at 10,000 students; trust lookup p50 / p95 | 11.9 s per sweep; 3.0 / 4.2 ms | 5 repetitions, run alone (exp8) |

Regimes and attacks are defined in Sections VI-C and VI-E. "Base" is the cycle search without the policy test; the deployed pipeline adds the fitted policy test. Hardware: Intel Core i7-12650H, 16 threads, 7.6 GiB for Docker.

TABLE V
CLOSURE VERSUS THE CLUSTERING COEFFICIENT ON CANONICAL NEIGHBOURHOODS

| Ego's neighbourhood | $C$ | $c$ | $\pi(C)$ | $\pi(c)$ |
|---|---|---|---|---|
| Sealed clique, 3 or 10 members | 1.00 | 1.00 | 0.70 | 0.70 |
| Clique of 4, one private friend each | 0.50 | 0.50 | 0 | 0 |
| Clique of 10, one private friend each | 0.80 | 0.80 | 0.39 | 0.39 |
| Clique of 10, two sociable friends each | 0.66 | 0.56 | 0.16 | 0.02 |
| Friends form a triangle, each with 20 outside ties | 1.00 | 0.05 | 0.70 | 0 |
| Hub, 8 unacquainted friends with one other tie each[†] | 0 | 0 | 0 | 0 |
| Sealed sparse group, 10 people with 4 ties each | 0.50 | 0.33 | 0 | 0 |

$C$: clustering coefficient; $c$: closure (4); $\pi(\cdot)$: structural penalty when that measure is used. [†]Counting ego inside its own neighbourhood gives $c = 0.50$.

with nothing dishonest about it. In the *coordinated* regime the rings are careful: before each farming errand the partner comes online and walks over, so the router offers the errand to the partner first. The *hard* regime combines honest favours with coordinated rings.

*Detectors.* Four single-signal baselines use structure or money at the production knees. A cycle baseline applies the ring search's leg rule and strongly-connected-component search to any pair of students, without the friendship test. Two graph-mining baselines come from the fraud literature. The first is FRAUDAR-style greedy peeling [33], [61] on the bipartite payer–payee graph, with edges weighted by the number of errands and the logarithmic column weighting FRAUDAR uses against camouflage. The second is label-propagation communities [28] on the friendship graph, ranked by the share of their members' value that stays inside. The last three rows are the ring search and the ring search followed by each version of the policy test.

*Single signals see nothing.* Ring members, like everyone, have friends outside their ring, so their closure stays under the 0.55 knee; and because farming errands are cheap and members keep running ordinary errands, circulation stays under 0.6. The clustering coefficient flags about five students per trial, mostly honest group members, and almost never a ring member.

*Graph-mining baselines find the marketplace, not the ring.* FRAUDAR's own output, its densest block, was the whole campus (1,199 or 1,200 accounts) in every trial. A ring of three to eight students who each run a dozen cheap errands for one another is not dense compared with the core of an active marketplace, where everyone pays everyone a little. Cut at the true number of ring members, the peeling order still reached an F1 score of only 0.28–0.48. Label-propagation communities were about the size of a hostel block, so no community crossed the circulation knee in any trial, and the ranked version mostly

TABLE VI
RING DETECTION ON A SIMULATED 1,200-STUDENT CAMPUS. PRECISION (P), RECALL (R) AND FALSE-POSITIVE RATE (FPR, %), MEAN OF 30 SEEDS

| | Routine | | | Busy | | | Favours | | | Coordinated | | | Hard | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Detector | P | R | FPR | P | R | FPR | P | R | FPR | P | R | FPR | P | R | FPR |
| Closure, circulation, or both | n/a | 0.00 | 0.00 | n/a | 0.00 | 0.03 | n/a | 0.00 | 0.00 | n/a | 0.00 | 0.00 | n/a | 0.00 | 0.00 |
| Clustering coefficient $> 0.55$ | 0.11 | 0.03 | 0.44 | 0.11 | 0.03 | 0.44 | 0.11 | 0.03 | 0.44 | 0.11 | 0.03 | 0.44 | 0.11 | 0.03 | 0.44 |
| Payment cycles, any pair | 0.63 | 0.95 | 1.34 | 0.38 | 0.96 | 3.73 | 0.50 | 0.94 | 2.23 | 0.69 | 0.99 | 1.12 | 0.48 | 0.99 | 2.48 |
| FRAUDAR-style peeling† | 0.36 | 0.36 | 1.41 | 0.28 | 0.28 | 1.59 | 0.30 | 0.30 | 1.55 | 0.48 | 0.48 | 1.16 | 0.43 | 0.43 | 1.27 |
| Label propagation + money† | 0.03 | 0.06 | 4.18 | 0.02 | 0.03 | 4.17 | 0.02 | 0.04 | 4.30 | 0.01 | 0.02 | 4.21 | 0.03 | 0.05 | 4.15 |
| Cycles among mutual friends (`find_rings`) | 0.98 | 0.95 | 0.05 | 0.89 | 0.96 | 0.29 | 0.82 | 0.94 | 0.50 | 0.99 | 0.99 | 0.03 | 0.83 | 0.99 | 0.47 |
| + policy test, router scores (original) | 0.99 | 0.95 | 0.02 | 0.98 | 0.96 | 0.05 | 0.94 | 0.94 | 0.13 | 0.10 | 0.02 | 0.00 | 0.15 | 0.04 | 0.17 |
| + policy test, fitted model (deployed) | 0.99 | 0.95 | 0.02 | 0.98 | 0.96 | 0.05 | 0.93 | 0.94 | 0.16 | 1.00 | 0.79 | 0.00 | 0.91 | 0.85 | 0.20 |

Recall is over the 26 planted ring members; FPR is over the 1,174 other students. Precision is counted as 0 in a trial that flags nobody; "n/a" marks detectors that flagged nobody in almost every trial. Structure is the same in every regime, so the clustering row repeats. †Scored as a ranking cut at the true number of ring members, which no deployment would know. Half-widths of 95% confidence intervals are at most 0.06 for precision or recall in the last three rows and at most 0.12 elsewhere; Table IV gives them for the deployed detector.

flagged a whole block of honest students. Both methods look for large dense or tightly-knit groups. The groups we are after are small, and dense only in money that flows around a cycle.

*Cycles need the friendship test.* Payment cycles without the friendship test found almost every ring member but flagged 13 to 44 honest students per trial, because repeated errands between nearby strangers also close cycles. Requiring every leg to join mutual friends is what makes cycle evidence usable: `find_rings` reached an F1 score of $0.96 \pm 0.02$ in the routine regime and $0.99 \pm 0.01$ against coordinated rings, which do not change the money pattern.

*The router's false positives.* The ring search's false positives come from the two cases it cannot see through: router-induced cycles (3.4 honest students per trial in the busy regime) and honest favours (5.9 in the favours regime, 5.5 in the hard one). Both versions of the policy test removed most of them. The fitted test removed 82% in the busy regime and 68% with honest favours, lifting F1 from 0.92 to 0.96 and from 0.87 to 0.93, and lost no ring member in either.

*Where the original test failed.* Against coordinated rings the router-score test cleared 146 of the 149 cycles that contained ring members, and recall fell from 0.99 to 0.02. The fitted test cleared 37 of them, for a recall of $0.79 \pm 0.06$ at a precision of 1.00. With honest favours added (the hard regime) the original test kept a recall of 0.04 and the fitted test 0.85, at a precision of 0.91. Section VI-D examines why.

*Variability.* Recall below 1 in the routine, busy and favours regimes came from seeds in which honest runners won enough of a ring's cheap errands to push one leg under the qualification floor (on average 1.3 of 26 members missed in the routine regime). Fig. 9(b) shows how that floor depends on farming intensity. The number of cycles raised per trial varied from four to nine, because some seeds plant honest groups where the router closes a cycle.

### D. Separating Routing From Choice

Fig. 7 plots the policy statistic (8) for every planted group in the main runs, under both acceptance models.

*Reading Fig. 7.* Each marker is one planted group in one trial. An open circle is an honest close-knit group and a filled square is a collusion ring. Each regime has one column: up to 840 honest groups (28 per seed over 30 seeds) and 150 rings (5 per seed). Markers are spread sideways only so that they do not hide one another; the horizontal position inside a column carries no meaning. The vertical axis is $z$, how many standard deviations a group's in-group takes lie above what the acceptance model predicted. At $z = 0$ the group did exactly what the model expected; a positive $z$ means its members took one another's errands more often than predicted. The horizontal line at $z = 3$ is the decision threshold. If the ring search flags a group below the line, the test clears it; a group above the line goes to a reviewer. The two numbers above each column count the honest groups (top) and rings (bottom) on or above the line. A good test puts the circles below the line and the squares above it. Panel (a) does this in the routine and busy regimes but not with coordinated rings, whose squares sit among the circles. In panel (b) those squares rise above the line while the circles stay near zero. In the favours and hard columns, many circles cross the line in both panels: honest groups that deliberately route favours look like rings to any test built on these logs.

*Calibration.* With the router's scores (Fig. 7(a)), honest groups in the routine regime had a median $z$ of $-2.1$: the router's scores predict more in-group takes than honest groups make, because every candidate in a tier is told at once and the order of the list matters little. The fitted model (Fig. 7(b)) centres honest groups near zero (median $-0.2$), which is what a calibrated test should do. Its weights describe the simulated behaviour as one would hope: averaged over seeds, a kilometre of distance cost 0.65 in log-odds and being a direct friend added 0.51, rising to 0.97 in the busy regime, where friends were simulated as four times likelier to accept. Friend-of-friend status added almost nothing ($-0.03$), and reputation is constant in the simulation, so its weight is zero.

*Separation.* In the routine and busy regimes both versions separated cleanly: every ring and at most 1% of honest groups reached $z = 3$. Against coordinated rings they part ways. With the router's scores only 3 of 150 rings reached the line (median $z = 0.4$), because a partner who is always the router's first pick looks, to a test that trusts the router, like the router's own doing. With the fitted model 112 of 150 did (median 3.7). A friend who takes a group's errands every time is unusual compared with how friends elsewhere on campus behave, and that comparison is what the fitted model makes.

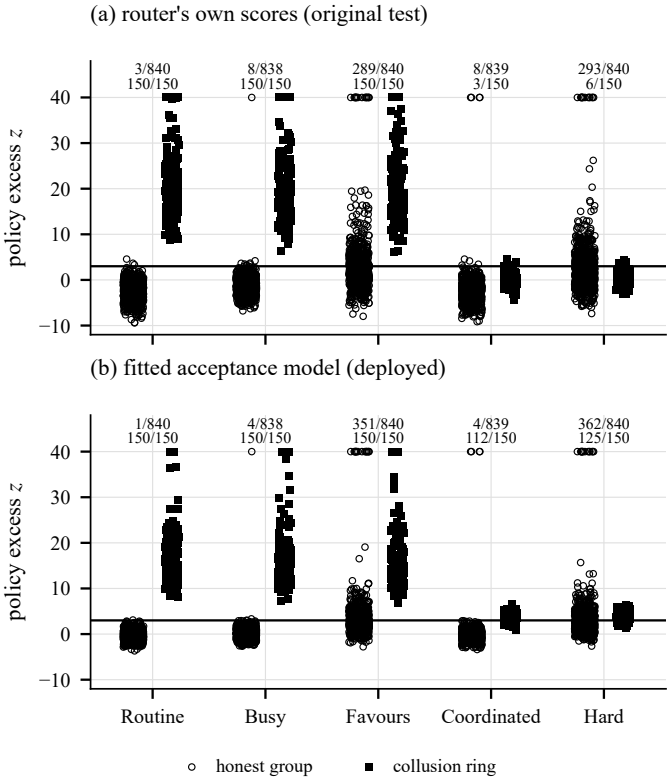



Fig. 7. Policy statistic $z$ for every planted group in the main runs (30 seeds per regime; 28 honest groups and 5 rings per seed) under (a) the router's own logged scores and (b) the fitted acceptance model. Numbers above each column count honest groups (top) and rings (bottom) with $z \geq 3$, which the test does not clear. Values above 40, including the $\pm 99$ of the degenerate case, are drawn at 40.

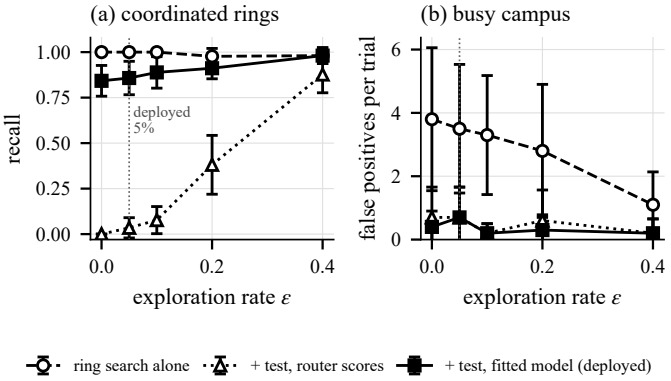



Fig. 8. Effect of the exploration rate. (a) Recall against coordinated rings. (b) Honest students falsely flagged per trial in the busy regime. Mean and 95% confidence interval over ten seeds; the dotted line marks the deployed rate.

*Honest favours.* When honest groups deliberately route favours to each other, 34% of them exceeded $z = 3$ under the router's scores and 42% under the fitted model, because the fitted model no longer starts from a conservative margin. The test only acts on groups the ring search raised, though, and of the 50 honest-only cycles raised in the favours regime it cleared 35 (the router-score version cleared 37).

*Exploration.* Fig. 8 varies the exploration rate, ten seeds per point. The router-score test needed exploration to see coordinated rings at all: its recall was 0.03 at the deployed 5%, 0.38 at 20% and 0.88 at 40%. The fitted test did not. Its recall was 0.84 with no exploration, 0.86 at 5% and 0.98 at 40%, because it learns from variation across the whole campus, not from the exploratory rounds of the group under test. In the busy regime both versions held false positives under one honest student per trial at every rate, against 1.1 to 3.8 for the ring search alone. Since exploration costs service (Section VI-G), the fitted model removes the main reason to raise it.

*Reading Fig. 8.* The horizontal axis of both panels is the exploration rate, the share of first offers sent without the social boost; the dotted vertical line marks the deployed 5%. Panel (a) measures recall against coordinated rings: the share of the 26 ring members still flagged after the policy test. The dashed line with open circles is the ring search alone, an upper bound, because the test can only remove flags, never add them. The dotted line with triangles is the original test, using the router's scores, and the solid line with filled squares is the deployed test, using the fitted model. The vertical gap between the dashed line and a test's line is the share of ring members that test wrongly cleared. The original test closes that gap only at 40% exploration; the fitted test stays close to it at every rate. Panel (b) counts honest students falsely flagged per trial in the busy regime, where the router itself puts honest friends together, so lower is better. Here the dashed line shows how many flags the router causes, and the two test lines show how many remain. Each point is the mean over ten seeds, and its error bar is the 95% confidence interval of that mean. The bars in panel (b) are wide because false positives are few and vary from seed to seed between zero and a handful.

*Trust weight.* The trust weight in (11) changed nothing about who accepted in our simulation: in-group shares and ring-search results were identical at 0, 1500 and 3000 m, because every runner in a tier is told at the same moment. It did change what the router-score test expected. With the weight at zero that test kept 97% of coordinated ring members flagged, and at 1500 or 3000 m it kept 3%. The fitted test kept 86% at all three weights, because it never reads the router's weights.

*The two limits.* Fig. 9 shows where relational evidence stops. As honest groups trade more favours (panel a), the ring search flags more of them: 0.1, 1.9, 7.8, 37.6 and 112.6 honest students per trial at 0, 3, 6, 12 and 24 favours per member. The fitted test cleared 63% of these at three favours per member and 76% at six, but only 27% at twelve and 2% at twenty-four. That much deliberate in-group routing is beyond what ordinary friends do, and the test is right to say so; what it cannot say is that the routing was innocent. Panel (b) shows the other end. Because a leg must carry at least five errands or Rs. 150, and honest runners sometimes take a ring's errand first, the ring search found almost no ring member when rings farmed four or fewer errands per member in a semester, 47% at eight and 97% at twelve. The *adaptive* attacker, a coordinated ring that chooses how much to farm, adds the policy test's own floor: against it the deployed pipeline caught 42% of members at eight errands, 86% at twelve and all of them at twenty-four. Below that floor the pipeline is blind, but the farming is

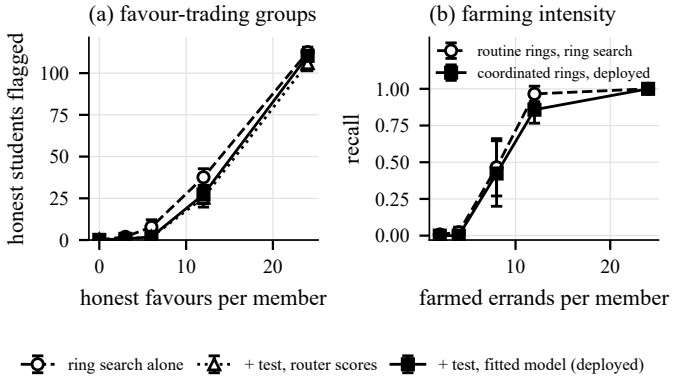



Fig. 9. (a) Honest students flagged as honest favour-trading grows (legend below). (b) Recall as rings farm fewer errands: careless rings against the ring search, and coordinated rings, which choose their volume, against the deployed pipeline. Mean and 95% confidence interval over ten seeds.

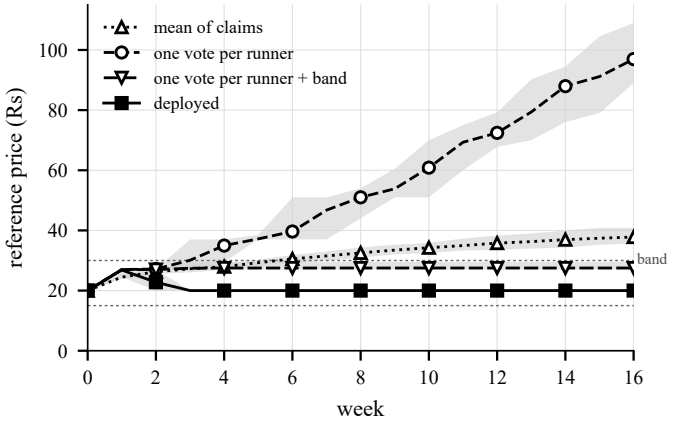



Fig. 10. Reference price week by week with 60% of runners line-walking. Lines are the mean over 30 seeds; shaded bands span the 5th to 95th percentile of seeds. The deployed estimator rises while the near-line rule is still collecting four claims from each attacker, then returns to the true price once their votes are excluded.

also small: at four errands per member the five rings together moved about Rs. 1,600 in a semester, and under one vote per rater a member farming from three partners gets three votes however many errands they run (Section VI-F).

*Reading Fig. 9.* In panel (a) the horizontal axis is the number of favours each honest group member deliberately routes to a friend in a semester, and the vertical axis is the number of honest students flagged per trial. The three lines use the same markers as Fig. 8. The gap between the dashed line (ring search alone) and the solid line (deployed test) is the honest flags that the test clears. That gap is wide at three and six favours, narrows at twelve and all but closes at twenty-four, where the lines meet: there the test, correctly by its own definition, finds the routing unexplained. In panel (b) the horizontal axis is the number of farmed errands per ring member in a semester, and the vertical axis is recall. The dashed line is careless (routine) rings against the ring search alone; where it rises shows the qualification floor of a payment leg. The solid line is coordinated rings, which choose their own volume, against the deployed pipeline; where it rises shows how much an adaptive ring can farm before it is caught. The point where a curve leaves zero is the volume a ring can stay under. Error bars are 95% confidence intervals over ten seeds; they are widest at eight errands, where whether a ring is caught depends on chance in individual seeds.

### *E. Reference Prices Under Attack*

*Setup.* We simulated 16 weeks of claims for one item, a chicken puff with an administrator reference of Rs. 20 and band Rs. 15–30, from 40 runners filing about 1.5 claims a week. Honest runners report Rs. 20, or Rs. 22 on the 15% of days the canteen charges more. Dishonest runners follow one of four strategies: *line*, claiming 95% of the allowance above the reference every time; *evasive*, doing so on a random half of their claims, intending to stay under the 60% share the near-line rule requires; *overt*, claiming Rs. 30; and *volume*, one runner claiming at the line ten times as often as anyone else. The reference is re-estimated weekly, and every claim is judged by the production code against the reference in force when it was made. Each configuration was run with 30 seeds, and we report the mean with the half-width of a 95% confidence interval.

*The reference.* Table VII shows three defences doing three different jobs. Excluding flagged claims stops overt inflation: the mean of all claims rose to Rs. 26.06 because flagged claims kept voting, while the mean of unflagged claims stayed near the true price. One vote per runner stops volume: a single runner filing ten times as often moved either mean by 12% and no median at all. Against line-walking, whose claims are never flagged, the vote limit is not enough on its own. The median-based estimators held with 30% dishonest runners and broke at 50%, and the one-vote estimator, whose allowance grows with the reference, ran away to Rs. 96.96 at 60% (Fig. 10). The band bounded that drift. What held the reference at Rs. 20 in every one of the 30 seeds was the exclusion of runners under a near-line flag. Across all 4,620 runs no honest runner received a near-line flag and no honest claim was judged FLAGGED.

*Reading Fig. 10.* The figure follows the hardest case in Table VII, 60% of runners line-walking, week by week. The horizontal axis is the week of the 16-week run, and the vertical axis is the reference price in force that week, against which every claim is judged. The true price is Rs. 20, and the two thin dashed lines mark the administrator's band of Rs. 15 to Rs. 30. Each line is the mean over the 30 seeds of one estimator, and the grey band around it spans the 5th to 95th percentile of seeds, so a narrow band means every seed behaved alike. A line that rises above Rs. 20 means the attack is moving the reference. That matters because the allowance grows with the reference, so each rise makes the next inflated claim look normal. The dotted line (mean of claims) drifts up steadily to about Rs. 38. The dashed line (one vote per runner) runs away to about Rs. 97, because the attackers are the majority of voters and their claims are never flagged. The band alone (long dashes) stops the drift at about Rs. 27.5, inside the band. The deployed estimator (solid line) rises to about Rs. 27 in the first two weeks while the near-line rule is still collecting four claims from each attacker. It then falls back to Rs. 20 and stays there once the flagged runners lose their vote.

TABLE VII
REFERENCE PRICE (RS.) AFTER 16 WEEKS, BY ESTIMATOR AND ATTACK. MEAN (95% CI HALF-WIDTH) OVER 30 SEEDS

| | Line-walking | | | Evasive | Overt | Volume |
|---|---|---|---|---|---|---|
| Dishonest runners | 30% | 50% | 60% | 60% | 60% | 1 of 40 |
| Mean, all claims | 24.14 (0.19) | 30.97 (0.40) | 37.80 (0.61) | 24.06 (0.18) | 26.06 (0.10) | 22.35 (0.12) |
| Mean, unflagged | 24.14 (0.19) | 30.97 (0.40) | 37.80 (0.61) | 24.06 (0.18) | 20.33 (0.02) | 22.35 (0.12) |
| Median, unflagged | 20.00 (0) | 31.27 (4.50) | 70.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) |
| One vote per runner | 20.00 (0) | 28.82 (0.97) | 96.96 (3.05) | 20.26 (0.26) | 20.00 (0) | 20.00 (0) |
| + line-walkers excluded | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) |
| + band only | 20.00 (0) | 28.54 (0.91) | 27.50 (0.38) | 20.26 (0.26) | 20.00 (0) | 20.00 (0) |
| Deployed (both) | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) | 20.00 (0) |

True price: Rs. 20, or Rs. 22 on dear days. "(0)" means every seed gave the same value.

TABLE VIII
MONEY AND TIME TO DETECTION UNDER THE DEPLOYED ESTIMATOR. MEAN (95% CI HALF-WIDTH) OVER 30 SEEDS

| | | Attackers | Days to | Excess paid (Rs.) | |
|---|---|---|---|---|---|
| Strategy | Attackers | flagged | first flag | before flag | 16 weeks |
| Line, 10% | 4 | 100% | 19.1 (2.0) | 110 (3) | 632 (20) |
| Line, 30% | 12 | 100% | 19.3 (1.2) | 331 (4) | 1958 (46) |
| Line, 60% | 24 | 100% | 19.0 (0.8) | 1016 (33) | 4370 (74) |
| Evasive, 10% | 4 | 85% | 48.0 (5.6) | 170 (12) | 316 (13) |
| Evasive, 30% | 12 | 79% | 46.4 (3.2) | 520 (21) | 954 (32) |
| Evasive, 60% | 24 | 83% | 45.3 (2.1) | 1019 (32) | 1986 (44) |
| Volume, 1 runner | 1 | 100% | 3.8 (0.7) | 27 (1) | 1591 (46) |
| Overt, 60% | 24 | n/a | n/a | −167 (8) | −167 (8) |

Excess paid is reimbursement above the true price, summed over all attackers. "Before flag" counts claims made before each attacker's first near-line flag; "16 weeks" assumes nobody reviews the flag. Overt claims are flagged one by one and reimbursed at the reference, so overt attackers recover less than they spend.

*Why most evasive attackers were caught.* An evasive attacker inflates half of their claims, so their long-run share of near-line claims is 0.50, and only 15–19% of them ever reached 60% over their whole history. Yet 79–85% were flagged. The reason is in how the rule is evaluated, not in its threshold. It runs after every claim, over the last 30 days, which for these runners holds about six judged claims. With six fair coin flips, four or more heads has probability $22/64 \approx 0.34$, and the rule gets a fresh chance at every submission. At the moment of flagging, the window held on average 6.4 claims, of which 4.5 were near the line, a share of 0.72. An attacker who wants to stay clear has to keep the *windowed* share under 60%, which at this claim rate means inflating well under half the time.

*Money and time.* Table VIII reports what the attacks cost. The near-line flag withholds nothing, by design, so the money only stops when an administrator acts on it. Line-walkers were flagged after about 19 days, having taken about Rs. 28 each by then. Evasive attackers who were flagged took about 46 days to reach it; counting the ones never flagged over their whole run, evasive attackers took about Rs. 43 each outside any flag. The 16-week column is what the platform would pay if nobody reviewed the flags. At 30% line-walkers it is Rs. 1,958, about Rs. 7 per inflated claim, which is why review capacity is part of the defence.

*Names.* Binding a claim to the order's lines (Section IV-B) was tested with 61 names a runner might type for three priced lines: misspellings, padded names ("chicken puff special"), obfuscated ones ("ch1cken puff", a Cyrillic letter, spaced letters), unrelated renames ("snack", "zzq special"), and swaps to the name of an unpriced line on the same order. Under the earlier rule, which matched names against the whole campus list, 41 of the 61 were paid unchecked. Under the deployed rule 20 are judged against the right reference, 35 are refused, and none is judged against the wrong one. The six that remain unchecked are the swaps. A runner can claim a priced item's money under the name of a hand-typed, unpriced line on the same order, and that line has no reference to judge it by. This is the cold-start case, and its bound is the escrow hold: a payout above the hold freezes for review (Section VI-G). The deterministic and fuzzy tiers also mapped 193 of 200 hand-written misspellings of 20 canteen items to the right key, none to a wrong one, and merged none of 20 look-alike items ("egg roll", "paneer tikka roll") into a listed one. We wrote all of these names ourselves, so the numbers measure coverage of the errors we anticipated, not accuracy in the field.

*Shops.* For shop-level pooling the same item was placed at a canteen charging the reference and at dearer ones. Against the campus reference every honest claim at a dearer counter is judged against Rs. 20, so an honest Rs. 30 claim is flagged and Rs. 10 withheld for telling the truth. Table IX(a) shows when (2) has moved the shop's reference far enough. A shop charging Rs. 30 needs three distinct reporting runners and one charging Rs. 40 needs ten. A shop charging Rs. 60 never gets there, because the drift cap holds its reference at Rs. 50; such an item needs an administrator to list it separately. The same mechanism is the weak point, and Table IX(b) measures it. The number of claims per colluder made no difference at 1, 10 or 50, because each runner has one vote. The number of colluders did. At a shop with few honest reporters, two or three colluders claiming Rs. 30 were paid in full, Rs. 10 above the true price per claim; with ten or twenty honest reporters they had to match them in number. Every colluding claim in the Rs. 30 rows still counted toward the walking-the-line rule, which is judged against the shop-adjusted reference, so colluders who keep doing it are flagged as line-walkers after four claims. Seeding the shop is not free either: until three distinct runners have reported there, each colluder's Rs. 30 claim is judged against Rs. 20 and flagged, which counts toward the strike ladder.

TABLE IX
SHOP-LEVEL POOLING AGAINST A CAMPUS REFERENCE OF RS. 20

(a) Distinct honest reporters before an honest claim is paid in full

| Shop price | Reporters needed | Shop reference then | Withheld before |
|---|---|---|---|
| Rs. 25 | 0 | 20.00 | 0 (within allowance) |
| Rs. 30 | 3 | 23.33 | Rs. 10 |
| Rs. 40 | 10 | 32.50 | Rs. 20 |
| Rs. 60 | never (cap Rs. 50) | 50.00 | Rs. 10 at 20 reporters |

(b) Colluders needed to be paid in full at a Rs. 20 shop

| Honest reporters | Claiming Rs. 30 | Claiming Rs. 40 | Claiming Rs. 50 |
|---|---|---|---|
| 0 | 3 | 10 | 20 |
| 1 | 2 | 8 | 20 |
| 3 | 3 | 8 | 15 |
| 5 | 5 | 8 | 15 |
| 10 | 10 | 11 | 11 |
| 20 | 20 | >20 | >20 |

Every colluder files 1, 10 or 50 claims; the counts are identical in all three cases. Honest reporters each claim Rs. 20 twice.

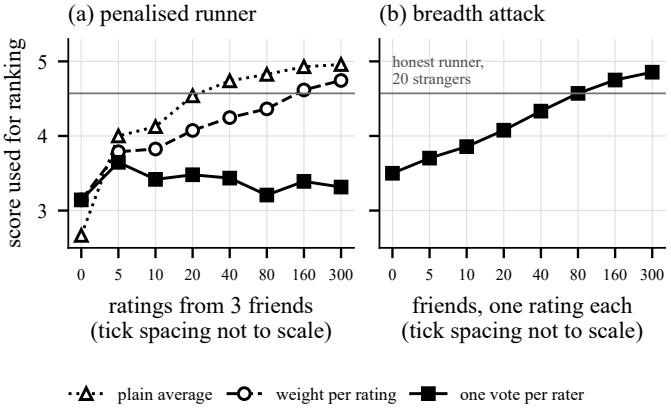



Fig. 11. (a) Ranking score of a penalised runner as three friends add five-star ratings, under three weightings. (b) The breadth attack under the deployed weighting, in which each rating comes from a different friend. The horizontal axes are categorical and roughly doubling, not linear. The histories are constructed, not sampled, so there is no seed variation to show.

### F. Reputation Farming

We built rating histories for several attackers and passed them through the production profile builder and farming detectors. The benchmark is an honest runner rated five stars by twenty strangers, whose ranking score is 4.57. An honest runner rated by twenty strangers (mostly five stars) and five friends scores 4.52, so a normal share of friends among one's customers costs nothing.

Fig. 11(a) follows a runner who served six strangers badly (mean 3.0), was penalised, and then collected five-star ratings from three friends. The plain average climbs to 4.96. Weighting each rating, which is how the system worked before it moved to one vote per rater, lets the farmed score pass the honest benchmark at 160 ratings. The deployed weighting stays between 3.21 and 3.65 however many ratings are farmed, because three friends remain three votes and their weight falls as their share of the history grows. With eight friends the deployed score peaks at 3.83. On the ranking scale of (11), this is the difference between the honest runner's 857 m advantage and at most 263 m for the farmer. The farming detectors flagged every history with 20 or more farmed ratings, whether the runner had been penalised or was farming from a standing start.

The breadth attack (Fig. 11(b)) is the limit. If every farmed rating comes from a different friend, one rating each, one vote per rater gives no protection: eighty such friends reproduce the honest benchmark of 4.57 exactly, and none of the three detectors fires because no friend rates twice and there is no stranger to compare against. The attack needs eighty real, verified accounts that have accepted a friend request and each paid for a real errand, and friendship maturity (Section IV-D) makes the new ties count for less in trust, though not in reputation. We do not defend against it beyond that cost.

*Reading Fig. 11.* The vertical axis of both panels is the score the dispatcher uses for ranking, $m$ in (11), on the one-to-five-star scale; the grey horizontal line at 4.57 is the honest benchmark. In panel (a) the horizontal axis is the number of five-star ratings the penalised runner has collected from three friends after the penalty. Its ticks (0, 5, 10, 20, 40, 80, 160, 300) are equally spaced although the values roughly double, so the axis is categorical, not linear. Three weightings of the same history are shown. The dotted line with triangles is the plain average that a student sees on a profile, and the dashed line with open circles weights every rating equally, as the system did before. Both climb past the benchmark, which means farming would pay. The solid line with filled squares is the deployed one-vote-per-rater score, which stays flat below 3.7 however many ratings are farmed. Panel (b) is the breadth attack under the deployed weighting. Its horizontal axis is the number of different friends who each give one rating, and the score reaches the benchmark at 80 such friends. Each point is one constructed history, run once through the production code; the computation is deterministic, so there are no error bars.

### G. Dispatch and Escrow

Table X dispatches the same 30 simulated semesters (routine regime, about 26,000 errands each) in four ways. Each runner's willingness and response delay for a given errand are fixed by a seed shared across policies, so the policies differ only in when each runner is told.

*Ranking alone does little.* Ranking a simultaneous broadcast by trust left 17.8% of errands with a runner within two hops. When every candidate is told at once, the order of the list cannot change who answers first. Staging the same offers in time doubled that to 35.8%, at a cost of 6.7 s in median wait and 42 s at the 90th percentile. Offering to one runner at a time in trust order reached 50.4% but more than doubled the median wait again.

*The router creates in-group circulation.* The same table measures the effect that motivates Section IV-F. Staging doubled the share of an honest group's errands that stayed inside the group, from 3.5% to 7.0%, and one-at-a-time dispatch raised it to 12.1%. No group member chose this; the router did. This is the circulation the ring detector reads, and it is why the detector must be conditioned on the router.

*The control group is not free.* The 5% coin is tossed for each logged first offer, and an errand whose friends-first round finds nobody is tossed again for the unrestricted round, so 7.0% of

TABLE X
DISPATCH POLICIES ON THE SAME SIMULATED SEMESTERS. MEAN (95% CI HALF-WIDTH) OVER 30 SEEDS

| | Broadcast, | Staged | | One at |
|---|---|---|---|---|
| | ranked | fixed | original | a time |
| Taken within 2 hops (%) | 17.8 (0.1) | 35.8 (0.2) | 36.2 (0.2) | 50.4 (0.3) |
| By a direct friend (%) | 5.1 (0.1) | 10.3 (0.1) | 10.4 (0.1) | 18.0 (0.1) |
| Honest group's errands kept inside (%) | 3.5 (0.1) | 7.0 (0.2) | 7.0 (0.2) | 12.1 (0.3) |
| Median wait (s) | 12.3 (0.0) | 19.0 (0.1) | 18.6 (0.1) | 41.9 (0.1) |
| 90th-percentile wait (s) | 36.7 (0.2) | 78.9 (0.1) | 76.8 (0.1) | 122.2 (0.4) |
| Unaccepted after 30 min (%) | 0.22 (0.01) | 0.22 (0.01) | 1.88 (0.03) | 0.22 (0.01) |
| unrestricted first offer (%) | n/a | 0.23 (0.02) | 4.38 (0.06) | n/a |

"Original" is the scheduler before the fix in Section V. About 40% of staged errands had no friend or friend-of-friend among the five nearest runners and went out unrestricted.

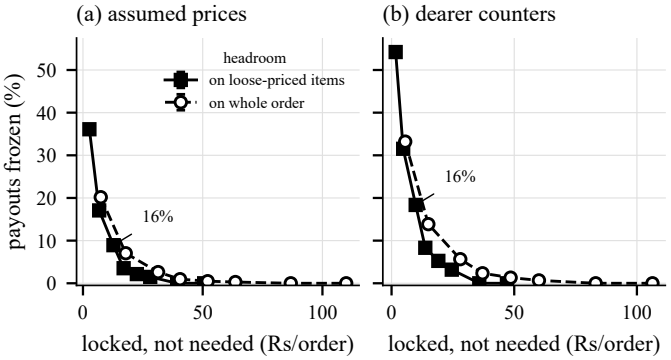



Fig. 12. Payouts frozen for review against money locked without need, as escrow headroom rises from 5% to 50%, for two choices of what the headroom covers, under (a) the assumed price model and (b) a dearer one. Error bars are 95% confidence intervals over 10 seeds of 3,000 orders.

errands went out exploratory. Of those, 12.8% went to a runner within two hops, against 37.5% in regular rounds.

*The scheduler fix.* When a requester has no friend or friend-of-friend among the five nearest runners, the first offer goes straight to those five. Under the original scheduler such an errand was never widened, and 4.4% of them went unaccepted for 30 minutes, which made staged dispatch the only policy with an overall expiry rate of 1.9%. With the 90-second widening applied to these errands too, their expiry fell to 0.23%, the same as the other policies, while the share taken within two hops moved only from 36.2% to 35.8%.

*Escrow.* We drew 3,000 orders per seed, mixing MRP goods with one to four loose-priced lines, and assumed that a loose item costs its listed price 70% of the time and 10%, 20% or 35% more otherwise. Each line was judged by the production judge and each hold computed by the production ledger code. With no headroom, 56% of payouts would freeze for review. Fig. 12 traces the trade-off as the headroom grows. For about the same money locked without need (Rs. 17–18 per order), headroom on loose-priced items alone froze 3.5% of payouts against 7.0% for headroom on the whole order, and full protection needed Rs. 39 against Rs. 87. The deployed 16% froze 9.0% (95% CI $\pm 0.2$) of payouts while locking Rs. 12.7 per order. Under a dearer price model, in which half of all loose items cost 10–60% more than listed, it froze 18.4%, and the advantage of targeting the uncertain part held along the whole curve. The same headroom bounds the cold-start case of Section IV-C: an unpriced line can be over-claimed only up to the slack in the hold, Rs. 12.7 per order on average and Rs. 32 at the 95th percentile here, before the payout freezes for a person to look at.

*Reading Fig. 12.* Each line traces one rule for what the escrow headroom covers. Each marker is one headroom percentage, from 5% at the left end of a line to 50% at the right end. The horizontal axis is the money a hold locks without need: the hold minus what the order finally paid, averaged per order. The vertical axis is the share of payouts that freeze for review because the runner is owed more than the hold. Both are costs, so a better rule has its curve nearer the bottom-left corner: fewer freezes for the same money locked. The solid line with filled squares puts headroom only on loose-priced items, as deployed, and the dashed line with open circles puts it on the whole order. The arrow marks the deployed 16%. In both panels the solid curve lies below the dashed one wherever the two lock the same money, which is the case for targeting the part of the order whose price is uncertain. Panel (b) repeats the experiment with dearer counters. Every point moves up, because more claims exceed any given hold, but the ordering of the two rules stays the same. Error bars are 95% confidence intervals over ten seeds of 3,000 orders; they are smaller than the markers at most points.

### H. System Measurements

*Cost.* Table XI times the evidence pipeline on synthetic campuses of 1,200 to 10,000 students with about 20 payments each, against one Neo4j 5.26 instance. Each graph job was run five times on the loaded graph and we report the median with the range; each trust lookup is one call of the production `social_scores` for five candidates, timed over 50 calls after 10 discarded warm-up calls. The experiment ran alone on the machine described in Section VI-A. At 10,000 students the three graph jobs together took 11.9 s, against a scheduler that runs them every ten minutes, and the trust lookup on the dispatch path took 3.0 ms at the median and 4.2 ms at the 95th percentile. The ring search now returns every ring, 117 of them on the largest synthetic campus, whose random payments close many cycles; this is what the earlier cap of 50 hid. The production Tarjan routine processed a million nodes and three million edges in 9.4 s, and fitting the acceptance model took 2.5 s on 10,000 logged rounds and 25 s on 100,000, about a semester's rounds for a campus of 5,000 students.

*Language channels.* The advisory channels play a small part in this design, and we tested them only for that part. On eleven hand-written cases with five runs each, the errand-text channel scored genuine groups 41.7 points higher than farmed ones on a 0–100 scale and cleared every honest control, but it also cleared one of three farmed groups whose errands had been written to look varied. The rating-comment channel showed no separation ($-1.2$ points) and failed to clear an honest runner whose friends rate without comments, so it is switched off. Neither channel followed instructions injected into the text. The cases were written by the authors, and the harness refuses to report an accuracy figure until twenty blind-labelled pilot

TABLE XI
COST OF THE EVIDENCE PIPELINE. MEDIAN (RANGE) OF FIVE REPETITIONS

| | | | Graph jobs (s) | | | Trust lookup |
|---|---|---|---|---|---|---|
| Students | Friendships | Payments | Closure | Circulation | Rings | p50 / p95 (ms) |
| 1,200 | 4,050 | 23,944 | 0.11 (0.06–0.14) | 0.81 (0.63–0.93) | 0.51 (0.50–0.54) | 5.7 / 14.8 |
| 2,500 | 8,408 | 49,895 | 0.14 (0.12–1.16) | 1.65 (1.57–2.90) | 1.25 (1.05–1.41) | 10.9 / 16.5 |
| 5,000 | 16,862 | 99,787 | 0.32 (0.26–0.40) | 3.19 (3.04–3.91) | 2.41 (2.20–2.63) | 4.0 / 6.3 |
| 10,000 | 33,755 | 199,525 | 0.54 (0.49–0.88) | 6.53 (6.13–7.27) | 4.91 (4.59–4.95) | 3.0 / 4.2 |

Tarjan routine alone, 3 out-edges per node: $10^4$ nodes 0.03 s; $10^5$ nodes 0.66 s; $10^6$ nodes 9.4 s
Acceptance-model fit: $10^3$ rounds 0.24 s; $10^4$ rounds 2.5 s; $10^5$ rounds 25 s

Hardware: Intel Core i7-12650H, 16 threads, 7.6 GiB available to Docker; Python 3.12; Neo4j 5.26 Community. Trust lookups: 50 timed calls after 10 warm-up calls, five candidates each.

cases exist, so these numbers show only that the errand-text channel works end to end, not that it is accurate.

*Tests.* The full suite of 273 automated tests, including at least one regression test for each of the six faults in Section V, passed in 111 s.

## VII. DISCUSSION AND LIMITATIONS

### A. What Relational Evidence Cannot Separate

A group of friends that deliberately routes favours to each other and a ring that farms ratings produce the same structure, the same money cycles and the same deliberate routing. Nothing in the friendship graph, the payment graph or the offer log distinguishes them; only intent does. This is an identifiability limit, not a tuning problem. The policy test asks whether a group took its own errands more often than friends elsewhere on campus do. Honest favour-trading answers yes for the same reason a ring does, and a careful ring that farms only as often as ordinary friends help each other answers no. Fig. 9 shows both ends: honest groups are flagged once favours become frequent, and the deployed test loses coordinated rings as they farm less. Only the content of the errands or a person can separate the two cases, and our language channel is not yet reliable enough to lean on (Section VI-H). We therefore keep a person between every ring flag and any sanction: ring and farming flags never feed the automatic ladder, and the automated parts of the system may demote and pause but never exclude.

### B. Why the First Policy Test Failed

The first version of the test used the router's own logged scores as its model of who would accept an errand. Within a tier every candidate is notified at once, so the order of the list has little effect on who answers first, and the scores overstated how much the router decides. A coordinated ring exploited exactly that. When the partner is always the router's first choice, a test that believes the router expects the partner to take the errand nearly every time and finds nothing surprising. Three results in Section VI-D point to the same cause. Exploration helped only slowly, setting the trust weight in the expectation to zero restored recall, and an acceptance model fitted to what runners actually did restored most of it at the deployed exploration rate, without that cost. The general lesson is that a detector conditioned on a matcher should condition on the matcher's measured effect, not on its design weights.

### C. Limitations

Every result here is from simulation. The friendship graph is a block model, not a real campus, and runner behaviour (willingness, response delay, friend affinity) is assumed. We varied the parameters that matter most, and the confidence intervals describe variation across seeds of that model, not uncertainty about whether the model is right. A pilot may find behaviour we did not imagine. The thresholds in Table III are reasoned defaults, not values fitted to data; the offer log exists partly so that a pilot can fit them. The acceptance model has four features, and a ring that varies distance and timing to look like an ordinary friend will be harder to separate than the coordinated rings simulated here. A ring whose members never declare friendship is invisible to every friendship-based mechanism, at the price of forgoing the trust boost; its cycles would appear only in the any-pair cycle search, which flags too many honest students to act on. The breadth attack defeats the reputation weighting at the cost of eighty real accounts. At a shop with few honest reporters, two or three colluders can move its reference, although the near-line rule still sees them. A runner can still claim a priced item's money under the name of an unpriced line on the same order, bounded only by the escrow hold. Line-walkers keep being paid the allowance until an administrator reviews the flag, so review capacity is part of the defence. Finally, the language-channel evaluation uses eleven hand-written cases.

### D. Ethics and Deployment

These mechanisms withhold money and can pause or suspend students, so we built them around three rules. Automated components may warn, demote and pause, but only a person may exclude: account suspension needs flags an administrator has upheld. Every flag carries the evidence that raised it, every sanction tells the student why, and a dismissed flag releases the money and lifts the pause it caused. And the advisory model runs on campus hardware, so that students' errand notes and reviews do not leave the institution. A pilot will collect data about identifiable students and will need institutional ethics approval and informed consent before it begins.

### E. Beyond Campus Errands

Nothing in the design is specific to food. It applies wherever expenses are reported rather than documented, such as shared

purchasing in hostels or offices and informal neighbourhood delivery, and more generally to any platform that uses social ties both to route work and to look for collusion. Once the router shapes who meets whom, the fraud detector must account for the router, or it ends up auditing the router's choices instead of people's.

## VIII. Conclusion

Remove the receipt from a delivery platform and every fraud control has to infer instead of verify. We built Errandly's controls around that fact. Price claims are bound to the lines the requester ordered and judged against a reference learned with one vote per runner inside a human-set band, and runners who habitually claim just under the limit lose their vote. Social closure discounts trust but never accuses; an accusation needs money cycling among mutual friends. And because the dispatcher's own preference for friends creates in-group circulation, ring flags are tested against an acceptance model fitted to how students outside every suspected group actually respond to offers.

In simulation the cycle search found rings that structure-only, money-only and graph-mining baselines missed, and the policy test removed most of the false flags that the router's preference for friends had caused. The test's first version failed in a way we did not anticipate: by borrowing the router's own weights as its model of behaviour, it cleared almost every coordinated ring. Fitting that model to logged acceptances repaired most of the failure without the service cost of more exploration. What no version can repair is the case in which honest friends and a careful ring route errands to each other in the same way; there, only a person who reads the errands can decide. The next step is a pilot with ethics approval, which can replace our assumed behaviour and hand-set thresholds with measured ones.

## Data and Code Availability

The datasets and the proof of every experiment are in the `paper-artifacts` folder of the project repository on GitHub:

https://github.com/chmm2/errandly/tree/feat/payments-fraud-detection/paper-artifacts

The folder holds the simulated datasets as CSV files, among them the campus, friendships, errands, payments and offer log of a representative seed, the price claims of each attack, the escrow orders and the item names. It also holds the raw output of every experiment with its seeds, the run logs, the result figures as images, the test report and a description of the machine. Its README maps every table and figure in this paper to the result file, dataset and script that produced it, and a checksum manifest lets a reader confirm the files are unchanged. The source code and the experiment scripts are in the same repository; the results were produced by the code of commit `c43fc91`, which the result files label `00d8928+review-fixes` because they were written before that commit was recorded. No data about real students were collected.

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## AUTHOR CONTRIBUTIONS

C. M. Mattam designed and built the backend services, the event pipeline, the mobile client and the social trust graph. U. Gogoi designed and built the escrow and payment flow and the price-claim fraud controls, and ran the evaluation. S. Sajlal built the web and administrator client, tested the applications, and contributed the literature review and the editing of the manuscript. R. S supervised the work and reviewed the manuscript. All authors approved the final version.

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